

1. To decrypt, you split y in half, obtaining y_1 and y_2 . Plug y_1 into F , then $\oplus$ the result with the first ciphertext. Plug y_2 into F , then $\oplus$ the result with the second ciphertext.
This is not secure. The attacker can know y , which is always the same. Any string $\oplus$ 'd with its complement is a string of 1s (therefore $r \oplus \bar{r}$ is all 1s).